

unit3

B.Sc. Hons. (CA&IT) / Data Science Sem-I
[U11E6CMT] COMPUTER MAINTENANCE AND
TROUBLESHOOTING

UNIT-3

**Software Maintenance and
Troubleshooting**

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1. Operating System (OS)

The **Operating System (OS)** is the core software that acts as a bridge between the user and the computer's hardware. It manages all the fundamental operations of a computer.

- **Key Functions of an OS:**
 - **File Management:** Organizes files and folders for easy access.
 - **Process Management:** Ensures programs run smoothly.
 - **Memory Management:** Allocates memory to various programs.
 - **Device Management:** Facilitates communication with hardware devices like printers and USB drives.
 - **Security:** Protects the system from unauthorized access and threats.
 - **User Interface:** Provides a way for users to interact with the computer, either through a Graphical User Interface (**GUI**) or a Command-Line Interface (**CLI**).
- **Examples:** Windows, macOS, Linux, Android, iOS.

2. File Management

Files is a collection of co-related information that is recorded in some format (such as text, pdf, docs, etc.) and is stored on various storage mediums such as flash drives, hard disk drives (HDD), magnetic tapes, optical disks, and tapes, etc

File management is the process of organizing, storing, and retrieving files efficiently. It involves arranging files in a **hierarchical structure** (directories and subdirectories) to prevent clutter and save time.

- **Importance:**
 - Saves time when searching for files.
 - Prevents accidental deletion.
 - Keeps data organized.
- **Tools:**
 - **Windows:** File Explorer
 - **macOS:** Finder
- **Key Functions:**

- **File Creation and Deletion:** Creating new files and removing unwanted ones.
- **Access Control:** The OS manages permissions to determine who can **read, write, or execute** a file using **Access Control Lists (ACLs)**.
- **Metadata Management:** The OS keeps track of file details like name, size, and creation date.
- **File Backup and Recovery:** Ensuring data safety by creating copies.
- **File Compression and Encryption:** Reduces file size and protects data from unauthorized access.(.zip, .rar, .7z files.)

Example:

In Windows, you can open **File Explorer** → Right-click → **New** → Create a new folder and name it.

3. System Updates

Updates are small software improvements released by OS developers to enhance performance and security.

- **Types:**
 - **Security Updates:** Protect the system from new viruses and hackers by patching vulnerabilities.
 - **Performance Updates:** Improve system speed and stability.
 - **Feature Updates:** Add new tools and functionalities.
 - **Driver Updates** – Keep hardware like printers, keyboards, and graphics cards working properly.
- **Importance:**
 - Protects from cyberattacks.
 - Improves efficiency and stability.
 - Ensures compatibility with new applications.

Example:

In Windows: **Settings** → **Update & Security** → **Windows Update** → **Check for updates**.

4. Security Settings

OS security is crucial to protect the computer from threats like viruses, malware, and unauthorized access.

- **Key Security Tools:**

- **Antivirus Software:** Detects and removes malware (e.g., Windows Defender, McAfee).
- **Firewall:** Keeps your computer safe by stopping unsafe internet connections.
- **User Accounts & Passwords:** Prevents unauthorized access to the system.
- **User Authentication:** Verifies user identity via passwords or biometric data.
- **Regular Audits:** Assessments to identify and fix security weaknesses.

Example:

In Windows: **Settings** → **Accounts** → **Sign-in options** to change your password or enable PIN/Face ID.

5. Recognizing & Addressing Common Software Issues

Even with a healthy OS, applications can sometimes misbehave.

5.1 Application Crashes

An **application crash** is when a **program suddenly closes** or becomes unresponsive.

- **Symptoms:**
 - "Program not responding" message.
 - The application window **disappears**.
 - The **program freezes**.
- **Causes:**
 - Software **bugs**.
 - **Insufficient memory** (RAM).
 - **Malware** infections.
 - **Conflicts with other programs**.
 - Outdated or **corrupt drivers**.
 - **Incompatible OS** updates.
- **Solutions:**
 - Restart the program.
 - **Restart the computer**.

- Update the application or OS.
- Reinstall the program.
- Scan for malware.
- **Troubleshooting Techniques:**
 - **Safe Mode:** a special way to start your computer that loads only the essential programs and drivers it needs to run.
 - **Clean Boot:** A troubleshooting method that lets you start your computer with a specific, limited set of startup programs and services. Unlike Safe Mode, which is very restrictive, a Clean Boot gives you more control.
 - **Device Manager:** The Device Manager is a control panel that lists all the hardware connected to your computer, like your keyboard, mouse, and graphics card. It lets you see if each piece of hardware is working correctly and allows you to update, disable, or uninstall its software, called a "driver."
 - **System File Checker (SFC):** A tool to detect and repair corrupted system files.

5.2 Outdated or Corrupt Drivers

Drivers connect OS and hardware.

If outdated/corrupt → apps may crash.

Fix:

- Open **Device Manager** → Update driver.
- Download latest driver from **manufacturer's website**.
- Use trusted **driver update tools**.

5.3 Insufficient System Resources (RAM/CPU/Disk)

Apps need enough memory & CPU to run.

Low resources → crash on startup.

Fix:

- Close background programs.
- Increase **virtual memory**.
- Upgrade **RAM/CPU** if needed.

5.4 Malware and Viruses

Malware corrupts files, eats resources, changes settings.

Can make apps unstable.

Fix:

- Run full **antivirus scan**.
- Use **malware removal tools**.

6. Safe Software Installation & Uninstallation

6.1 Safe Installation

- **Where to Download:** Always **download** software **from trusted sources** like **official websites** or dedicated App Stores. Avoid third-party torrent sites or pop-up ads.
- **Steps:**
 - **Download from trusted sources**
 - Use official websites or verified app stores (e.g., Microsoft Store, Google Play, Apple App Store).
 - Avoid downloading from unknown or suspicious websites.
 - **Check system requirements**
 - **Make sure your computer meets the minimum hardware and OS requirements** for the software.
 - **Scan for viruses**
 - Before opening the installer file, scan it with **antivirus software**.
 - **Read installation prompts carefully**
 - Avoid clicking “Next” quickly.
 - **Deselect any unwanted extra software** or toolbars.
 - **Choose installation type wisely**
 - **Use Custom Installation** to **control what gets installed**.
 - Avoid “express” installation if it installs unnecessary programs.
 - **Keep software updated**

- **Enable automatic updates** for better performance and security.

Example:

When installing Zoom, download it from **zoom.us** (official site) instead of third-party download sites.

6.2 Proper Uninstallation

Simply deleting a program's folder doesn't fully uninstall it. It leaves behind junk files and registry entries that can waste storage and cause errors.

- **Correct Methods:**
 - **Use official uninstall methods**
 - In Windows: **Control Panel → Programs → Uninstall a program.**
 - In macOS: Drag the application to the Trash, then empty Trash.
 - **Remove leftover files**
 - After uninstalling, **check** for leftover folders in “**Program Files**” or “**Documents**”.
 - **Avoid deleting system files**
 - **Do not manually delete files from the OS folder** (like Windows or System32).
 - **Backup important data**
 - If the software stores files (like **project files or settings**), back them up before uninstalling.
 - **Restart after uninstalling**
 - Helps the system **clear temporary files and free resources.**

7. Basic System Recovery Tools

- Sometimes a computer may **stop working properly** due to **software errors, viruses, driver issues, or system file corruption.**
- **System recovery tools** help restore the computer to a working state without needing a complete reinstall.

7.1 System Restore (Windows)

System Restore is a Windows feature that reverts your computer's system files and settings to an earlier date.

- **When to Use:** After installing faulty software or a driver.
- **Control Panel → Recovery → Open System Restore**
- **Effect:** It only affects system settings; your personal documents and files remain untouched. (Does not affect personal files like photos or documents.)

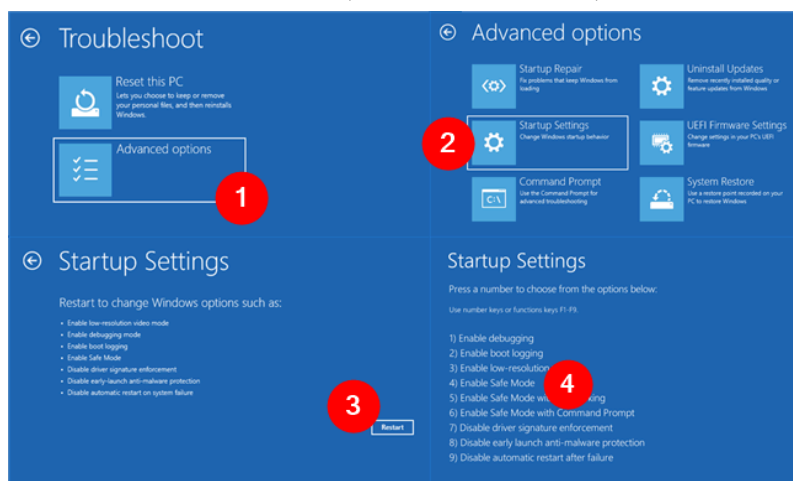
7.2 Backup & Restore

Backup is the process of creating copies of important files to protect against data loss.

- **Importance:** Helps recover data after a crash, virus attack, or accidental deletion.
- **Methods:**
 - **External Drives:** Storing data on an external hard drive or a pen drive.
 - **Cloud Storage:** Backing up data to online services like **Google Drive**, **OneDrive**, or **iCloud**.

7.3 Safe Mode

- **Purpose:** Starts Windows with only essential programs and drivers.
- **When to Use:** To fix problems caused by faulty drivers, software, or viruses.
- **How to Access (Windows 10/11):**

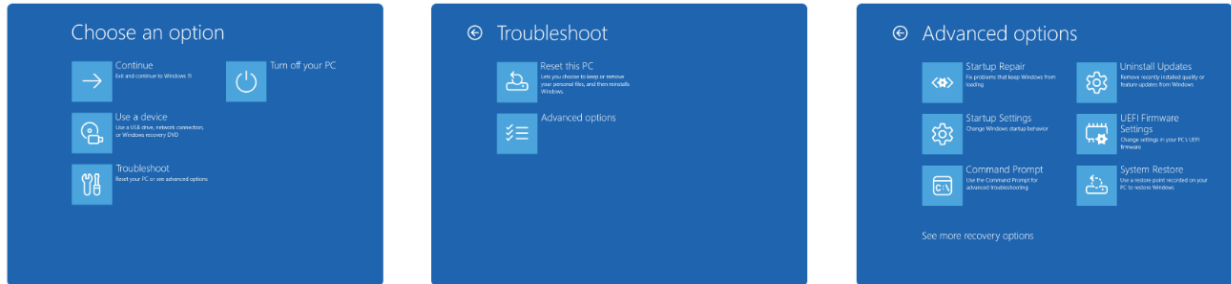


Unit-3: -Software Maintenance and Troubleshooting:

- Hold Shift while clicking
- **Restart → Troubleshoot → Advanced options → Startup Settings → Enable Safe Mode**

7.4 Startup Repair

- **Purpose:** Fixes problems that stop Windows from starting properly.
- **How to Access:**



- Boot from **Windows installation media → Choose Repair your computer → Troubleshoot → Advanced options → Startup Repair.**

7.5 Reset This PC

- **Purpose:** Reinstalls Windows to fix major problems.
- **Options:**
 - Keep my files – Removes apps and settings but keeps personal files.
 - Remove everything – Deletes all files and reinstalls Windows.
- **How to Access:**
 - **Settings → Update & Security → Recovery → Reset this PC.**

Tip for Students:

- Always try System Restore or Safe Mode first before doing a full reset, because resetting will remove your apps. And remember to keep regular backups to avoid losing important work.

❖ Hands-on Session: Practicing Software Troubleshooting & System Recovery

Part 1: Practicing Software Troubleshooting

Activity 1: Checking for Slow System Performance

1. Press Ctrl + Shift + Esc → Open Task Manager.
2. Go to the Processes tab → Check which app is using more CPU/RAM.
3. Right-click the heavy app → Select End Task.
👉 Observation: System becomes faster.

Activity 2: Fixing Application Not Responding

1. Open a program (e.g., Word) → Force it to stop responding (open too many files).
2. Use Task Manager → End Task to close it.
3. Restart the program.
👉 Observation: Program works again.

Activity 3: Checking Internet Issues

1. Disconnect from Wi-Fi → Try opening a website.
2. Reconnect Wi-Fi → Check again.
3. If still not working:
 - o Open Command Prompt → type ping google.com.
 - o If no reply → Check router or network settings.👉 Observation: Students see how to test connectivity.

Activity 4: Fixing Startup Problems

1. Press Windows Key + R → type [msconfig](#).
2. Go to Startup tab → Disable unnecessary programs.
3. Restart system → Check faster boot time.
👉 Observation: Startup is quicker.

Part 2: Practicing Basic System Recovery

Activity 5: Using System Restore

1. Press Windows Key + S → type System Restore.

2. Choose a restore point (before a problem occurred).
3. Follow the steps → System rolls back to earlier working state.
👉 Observation: Older settings/software are restored.

Activity 6: Booting in Safe Mode

1. Restart computer → Press F8 / Shift + Restart.
2. Select Troubleshoot → Advanced Options → Startup Settings → Safe Mode.
3. System starts with only essential programs.
👉 Observation: Safe Mode helps fix driver/software issues.

Activity 7: Basic Virus Scan (Windows Defender)

1. Press Windows Key + S → type Windows Security.
2. Select Virus & Threat Protection → Quick Scan.
3. Remove any detected threats.
👉 Observation: Students see real-time protection in action.

Activity 8: Data Backup & Recovery

1. Copy a file to USB / Cloud (Google Drive).
2. Delete original file from computer.
3. Restore it from backup.
👉 Observation: Backup protects from data loss.